

STEPCLIMBER · CALCULATION NOTE

StepClimber sizing calculations

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DOCUMENT CONTROL

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REVISION HISTORY

Version	Date	Author	Change
0.3	2026-10-01	Amish Chadha	Recalculated for the constructable design (SCM-DDR-003); mass itemised, chain centres to whole chains, bearings inside the frame, frame-fixed outlines checked against the nosings; budget reported as a value-engineering target
0.2	2026-09-25	Amish Chadha	Recommendations accepted by Amish (DDR-002); recalculated for 200 mm wheels, 150 mm arms, a two-stage chain drive, a plus or minus 6 degree tilt window and the revised R3, R5 and R12 targets
0.1	2026-09-25	Amish Chadha	First TRL 3 sizing note (cluster geometry, nosing clearances, torque, energy, holding, handle force, structure, mass, size, cost) with a status for every requirement

StepClimber sizing calculations

This version recalculates StepClimber for the constructable design of SCM-DDR-003, made on 2026-10-01 under Amish's 2026-09-30 instruction to make the design physically buildable. The cluster, ratios, tilt window and stair geometry of SCM-DDR-002 are unchanged. What changed is that every part now exists and is fixed to its neighbours: the bearings sit inside the frame on welded axle plates, the countershaft runs in two bearings in a closed chain case, the chain centres suit whole chains, and the parts on the frame back hang on uprights. Naming every part also lets the mass be added up rather than allowed for, and the truck comes out at 34.2 kg (26.3 kg in v0.2).

Three requirements are not met. R5: 34.2 kg against 27 kg. R3: at the new mass the climb needs 260 W at 17 steps/min from the 250 W motor, which climbs at 16.3 steps/min. R6: the grip force at the plus or minus 6 degree window edge is 121 N against 100 N. Seven are met on paper and four cannot be verified without hardware. Every frame-fixed part near the shaft clears every nosing in the R2 range by 10 mm or more (R15), the safety factors at a 3 g dropped-step load are 1.4 or more, and the pack gives about 1,215 loaded steps per charge. The estimated cost is USD 776 against the USD 650 value-engineering target.

Every number here is printed by `docs/04-calcs/sizing.py` (run from the repo root: `python docs/04-calcs/sizing.py`), which also writes `docs/04-calcs/results.csv`. The script reads the geometry from `cad/src/model.py` and the costs from `bom/bom.csv`. All values are first-principles estimates for a paper design; nothing is measured.

■ 1. Assumptions

Table 1. Inputs. All are assumptions for a paper design.

Input	Value	Basis
Payload	60 kg (132 lb)	R1, decided 2026-09-25 (SCM-DDR-001)
Truck mass	34.2 kg (roll-up in section 9)	Component estimates and the constructable model
Combined center of mass	500 mm above the shaft, along the frame	Parcels stacked on the toe plate
Design stair	196 mm (7.75 in) risers, 254 mm (10 in) treads; nosing overhang 0 to 32 mm	IRC limits; R2
Climb speed	17 steps/min: one third of a turn in 3.5 s	R3, revised 2026-09-25 (SCM-DDR-002)
Tilt window	Plus or minus 6 degrees about the set angle	R7, decided 2026-09-25 (SCM-DDR-002)
Drive efficiencies	Driver 95 %, brushed motor 80 %, self-locking worm 45 %, first chain stage and bearings 95 %, second chain stage 97 %	Typical parts
Worm	Single start, 4 degree lead, ratio about 75	Typical wheelchair worm gearmotor, 40 rpm output
Brake	2.0 N·m spring-applied on the motor shaft	Typical wheelchair motor brake
Dynamics factor	1.3 on static shaft torque	Friction and the landing thump
Dropped-step shock	3 g on the structure	A cluster dropping off a nosing
Materials	Shaft 1018 cold drawn, 370 MPa yield; spider plate S275; frame tube 250 MPa after welding	Common stock
Chain	06B: 9.525 mm pitch, 8.9 kN minimum tensile; 08B: 12.7 mm, 17.8 kN	ISO 606 catalog minimums, to confirm with the supplier
Pack	256 Wh LiFePO4, 90 % usable, 10 % extra for standby, starts and stops	R13
Charger	29.2 V, 3 A, 90 % efficient, plus 0.3 h constant-voltage tail	R13
Rolling resistance	0.03	Solid rubber on a smooth floor
Operator	Stands uphill of the truck and pushes or pulls the grip horizontally while climbing	Normal hand truck practice on stairs

The design load is 94.2 kg in total, a weight of 924 N. The clusters are assumed to carry all of it while lifting, which is conservative.

■ 2. Cluster geometry (R2)

The cluster has three 200 mm wheels on 150 mm arms, so the wheel centers are 259.8 mm apart, the shaft sits 175.0 mm above the floor and the cluster is 500 mm across.

A climb starts with the wheel nearest the stair lodged against the riser. The cluster turns about that wheel (the pivot) until the top wheel lands on the tread above, 259.8 mm from the pivot and one riser higher. The landing wheel's contact point is therefore $\sqrt{d^2 \text{ minus } h^2}$ minus 100 mm past the riser face, plus the nosing overhang. Table 2 shows how far it lands past the nosing edge.

Table 2. Landing point of the swinging wheel past the nosing edge.

Riser	No overhang	32 mm overhang	Tread needed to land
100 mm	139.8 mm	139.8 mm	239.8 mm
150 mm	112.1 mm	125.5 mm	212.1 mm
175 mm	92.0 mm	124.0 mm	192.0 mm
196 mm	70.5 mm	102.5 mm	170.5 mm
200 mm	65.8 mm	97.8 mm	165.8 mm
210 mm	53.0 mm	85.0 mm	153.0 mm
220 mm	38.2 mm	70.2 mm	138.2 mm
241 mm	Does not land on a square nosing (minus 2.9 mm)	29.1 mm	

With at least 30 mm of landing past a square nosing, the largest riser the cluster climbs is 225 mm (209 mm on the v0.1 cluster), which covers the 100 to 200 mm range of R2 with room for steeper older stairs. The longest tread needed is 239.8 mm, on low risers, so every tread of 250 mm or more is long enough; this is the one geometric cost of the longer arm.

After the landing, the cluster turns on about the landed wheel for the rest of the third of a turn, which lifts the old pivot wheel clear, and the courier rolls the truck back on the landed wheel to the next riser.

■ 3. Nosing clearances (R2, R15)

The script steps the cluster through a full climb in 2D and records how close the hub, the spider arms and the wheels come to the nosing corners.

Table 3. Closest approach to a nosing during one step.

Riser	Overhang	Hub to nosing	Spider arm clearance	Wheel clearance
100 mm	0	119.6 mm	57.0 mm	0.0 mm
100 mm	32 mm	119.6 mm	57.0 mm	0.0 mm
125 mm	0	105.6 mm	50.0 mm	3.1 mm
125 mm	32 mm	105.0 mm	47.3 mm	0.0 mm
150 mm	0	93.9 mm	47.4 mm	11.8 mm
150 mm	32 mm	89.5 mm	35.2 mm	0.0 mm
175 mm	0	87.1 mm	50.0 mm	25.0 mm
175 mm	32 mm	71.6 mm	19.5 mm	0.5 mm
196 mm	0	87.8 mm	49.1 mm	22.4 mm
196 mm	32 mm	64.6 mm	25.9 mm	16.6 mm
200 mm	0	88.8 mm	48.4 mm	19.7 mm
200 mm	32 mm	64.2 mm	27.7 mm	19.9 mm

The nosing passes through the gap between the two lower arms, close to the shaft line, at the moment the swinging wheel lands. Anything on the shaft line must therefore stay within the minimum hub-to-nosing distance less a 10 mm margin: **77 mm with square nosings and 54.2 mm with 32 mm overhangs** (48 and 26 mm on the v0.1 cluster).

- The 25 mm shaft, the 38 mm weld-on hubs, the 48 mm spider centre discs and the 45 mm flange bearing housings clear every nosing in the R2 range.
- **The 08B 20-tooth shaft sprocket (88 mm across) and its guard sweep a 53.9 mm radius, 0.3 mm inside the 54.2 mm limit**, so the 10 mm clearance is kept with 32 mm overhangs. R15 is met, with the guard sized to the limit; a thinner guard gap would add margin.
- The countershaft sits 143.9 mm up the frame from the cluster shaft and 15 mm toward the stair (SCM-DDR-003, P5 and P6). Its 35-tooth sprocket guard (65.8 mm radius) passes no closer than 146 mm to any nosing, a clearance of 80 mm.
- The constructable design adds parts on the frame near the shaft. The script turns every nosing position of the climb into the frame's own coordinates and measures the distance to each outline (Table 3a). All clear by 10 mm or more.
- **The spider arms clear nosing overhangs of 32 mm** in the R2 riser range, with 19.5 mm to spare at the worst case (175 mm riser). R2 is met. The wheels touching a nosing briefly (clearance near zero) is acceptable: they are rubber and roll over it.
- The frame back clears the nosings above the countershaft by 109 mm beyond the skid face at the 30 degree climbing tilt.

Table 3a. Frame-fixed outlines near the shaft against every nosing in the R2 range (10 mm needed).

Outline	Nearest nosing
Chain case and drive-side axle plate (53.9 mm on the shaft line, 65.8 mm round the countershaft)	10.3 mm
Plain-side axle plate, 92 x 144 mm	13.5 mm
Main flange bearing	27.2 mm
Countershaft flange bearing	113.6 mm
Gearmotor gearbox	118.5 mm
Motor and brake	127.2 mm
Replacement low cross bar	96.6 mm
Rails, load side of the shaft	79.5 mm
Nosing guard skids (meant to be the nearest part)	28.4 mm

Table 4. Cluster sizes compared at v0.1 (hub to nosing and arm clearance with square and 32 mm nosings). The 150 mm arm with 200 mm wheels was recommended and decided.

Arm	Wheel	Hub to nosing	Arm clearance	Landing at 200 mm riser	Tread needed at 100 mm	Hub height
135 mm	150 mm	58 / 36 mm	21 / -8 mm	46 mm	211 mm	142 mm
135 mm	200 mm	93 / 70 mm	54 / 29 mm	21 mm	211 mm	168 mm
150 mm	150 mm	52 / 29 mm	14 / -15 mm	91 mm	240 mm	150 mm
150 mm	200 mm	87 / 64 mm	47 / 20 mm	66 mm	240 mm	175 mm
165 mm	200 mm	81 / 64 mm	42 / 12 mm	104 mm	268 mm	182 mm

The 54 mm envelope takes an 08B 20-tooth sprocket (chain safety factor 4.0, section 4) or a 06B 27-tooth one (safety factor 2.2, too low). Because the final stage is then only 2:1, the drive has a second chain stage: a 06B 10-tooth sprocket on the gearmotor drives a 35-tooth sprocket on the countershaft (3.5:1), for 7:1 overall.

■ 4. Torque, speed and power (R3)

At 17 steps/min the shaft turns at 0.593 rad/s (5.67 rpm). Lifting 924 N through 196 mm takes 181 J per step, a mean shaft torque of 86 N·m over a third of a turn.

The drive torque at any instant equals the vertical load times the horizontal distance from the support wheel to the hub, plus the grip force times the vertical distance. The peak is 140 N·m static, when the hub passes level with the landed wheel (the full 150 mm arm), and 182 N·m with the 1.3 dynamics factor. That needs 108 W at the shaft and **260 W from the motor**, 4 % over its 250 W rating. **R3 is not met**: the fastest climb this motor supports at the new mass is 16.3 steps/min (17.8 in v0.2). Options are proposed in SCM-DDR-003, item A2.

Table 5. Two-stage chain drive at the peak torque.

Stage	Sprockets and chain	Ratio	Centres	Chain pull	Safety factor on minimum tensile
First, gearmotor to countershaft	06B, 10T to 35T, 54 links	3.5:1	145.1 mm	1,764 N (94 N·m at the countershaft)	5.0
Final, countershaft to cluster shaft	08B, 10T to 20T, 38 links	2:1	144.7 mm	4,478 N	4.0

The centres are set so that each chain is a whole number of links (even counts, so no offset link); slots let the countershaft bearings and the gearmotor move 4 mm either way to tension them. The 7:1 drive needs the gearmotor to give 40 rpm and 28.2 N·m, inside its 30 N·m rating.

■ 5. Energy and endurance (R4, R13)

Table 6. Energy per loaded step on the design stair.

Quantity	Value	Basis
Drive efficiency up	31.5 %	$0.95 \times 0.80 \times 0.45 \times 0.95 \times 0.97$
Pack energy per step up	575 J (0.16 Wh)	181 J / 0.315
Pack energy per step down	46 J	Lowering through a self-locking worm
Usable pack energy	829 kJ	256 Wh x 90 %
Loaded steps per charge, up and down	1,215	$829 \text{ kJ} / ((575 + 46) \text{ J} \times 1.1)$
Typical day	25 drops of three floors (48 steps)	
Charge time	3.5 h	$256 \text{ Wh} / (29.2 \text{ V} \times 3 \text{ A} \times 0.9)$ plus 0.3 h

Lowering: a 45 % efficient worm with a 4 degree lead has a friction angle of 4.83 degrees. Driving it while the load pushes needs $\tan(4.83 \text{ minus } 4) / \tan 4 = 21 \%$ of the lift work at the worm, so the motor supplies 46 J per step down instead of braking.

The energy flow for one step up is 575 J from the pack, 546 J after the driver, 437 J at the motor shaft, 197 J after the worm and 181 J of lift (Figure 2 of SCM-PRC-001). R4 (1,000 steps each way) and R13 (4 h charge) are met.

■ 6. Holding on the stair (R8)

The shaft must hold 140 N·m at the worst point of a step, or 20.0 N·m at the gearmotor output.

- **Worm.** Its back-drive efficiency is $2 \text{ minus } 1/0.45 = -0.22$. A value at or below zero means the worm cannot be driven backwards by the load when static, so it holds without power. Vibration can lower the friction and let a self-locking worm creep; this cannot be checked on paper.
- **Brake.** If the worm were fully back-drivable, the motor brake would need $140 / (75 \times 7) = 0.27 \text{ N·m}$. A 2.0 N·m brake gives a 7.5-times margin.

Two independent holds exist on paper. The drift limit in R8 (5 mm in 10 minutes) needs a test and is not verifiable at TRL 3.

■ 7. Balance and handle force (R6, R7, R10)

During a step the hub moves relative to the wheel carrying the load: from 29 mm on the stair side of the support wheel to 150 mm on the far side. The frame angle is held, so the center of mass moves with the hub. The best set angle puts the center of mass 61 mm on the stair side of the hub, which leaves a swing of plus or minus 89 mm about the support point. At the edge of the plus or minus 6 degree window the center of mass moves a further 52 mm.

Table 7. Grip force during a climb on the design stair.

Case	Grip force
At the set angle, worst point of the step	76 N
At the plus or minus 6 degree window edge	121 N (R6 not met)
Widest window that meets 100 N	Plus or minus 3.1 degrees (99 N)
Vertical grip force at the window edge (flat hand truck convention)	273 N

The horizontal-force case is the relevant one: the courier stands uphill and pushes or pulls the grip, which is about 1.1 m above the support wheel. **R6 is not met by 21 %**, up from 11 % in v0.2 because the truck is heavier. The options are set out as SCM-DDR-002 item 14 and SCM-DDR-003 item A3, proposed, awaiting Amish.

Pushing on the flat takes $0.03 \times 924 \text{ N} = 28 \text{ N}$, which meets R10.

■ 8. Structure (R1)

Table 8. Stresses at the peak torque and a 3 g dropped-step load.

Part	Load	Stress	Safety factor on yield
Shaft, 25 mm	182 N·m torsion (59 MPa); bending over the 93 mm from the bearing centre to the wheel plane (84 MPa)	133 MPa von Mises; keyway factor 2	1.4
Spider arm, 45 x 6 mm, 150 mm long	1.39 kN at the wheel: bending 103 MPa; twist from the 30 mm wheel offset 84 MPa	178 MPa von Mises	1.5
Frame rail, 28 x 1.5 mm tube	76 N·m each from the window-edge grip force	97 MPa	2.6

R1 is met statically. The shaft margin is lower than in v0.2 (1.8) because the bearings now sit inside the frame, where they can be fixed, so the shaft overhangs its bearing by 93 mm rather than 48 mm (SCM-DDR-003, P1). Fatigue of the keyed shaft and the welded frame over many thousands of steps is not assessed here and belongs with a later test.

■ 9. Mass and size (R5, R11)

Table 9. Mass roll-up. Lines 14 to 17 and the stub axles were added to make the design buildable (SCM-DDR-003); line 3 is now itemised.

Item	Mass
1 Frame and toe plate (the replacement low bar replaces the bar cut off)	8.0 kg
2 Clusters (pair), 200 mm wheels	5.5 kg
2 Stub axles (6), 20 mm bar	0.8 kg
3 Cluster shaft 1.9 kg, countershaft 0.3 kg, four flange bearings 2.1 kg, four sprockets 0.7 kg, two chains 0.6 kg	5.5 kg
4 Worm gearmotor with brake	4.5 kg
5 and 6 Driver, controller, IMU	0.8 kg
7 Pack and cradle	2.6 kg
8 Switch, fuse, harness	0.5 kg
9 Handle controls	0.6 kg
10 Strap	0.4 kg
11 Skids	0.4 kg
13 Enclosure and hardware	0.5 kg
14 Axle plate 0.4 kg, case outer plate 1.1 kg, case inner plate 0.4 kg, band 0.3 kg, spacers 0.3 kg	2.5 kg
15 Component uprights	0.6 kg
16 Skid standoffs	0.5 kg
17 Fixings added for construction	0.5 kg
Total	34.2 kg

R5 (27 kg or less) is not met, by 7.2 kg. About 3.0 kg of the rise comes from itemising the drive, which v0.2 carried as a 2.5 kg allowance (the 25 mm shaft alone is 1.9 kg), and about 4.4 kg from the parts and fixings added to make the design buildable. The case plates are already thin steel and aluminium; options are proposed in SCM-DDR-003, item A1. The pack is 2.6 kg (3 kg or less, met).

From the model, the truck is 569 mm wide at the wheel faces and 588 mm over the wheel end screws, and 1,451 mm tall upright, which meets R11 (600 and 1,500 mm). A folding hinge 950 mm above the shaft (the Option C study, not adopted) would fold the truck to 1,139 mm.

■ 10. Cost (R12)

Value-engineering target: USD 650 (`budget_usd` , a hypothetical control target, not a limit). Estimated cost of the constructable design: USD 776, from the 17 BOM lines (USD 126 over the target). The parts added for construction (lines 14 to 17) cost USD 99, and the weld-on hubs, the extra bearing and the connecting links USD 44 more on lines 2 and 3. The main cost drivers and the savings worth trying are listed in the design decisions register (SCM-DEC-001).

■ 11. Requirement status

Table 10. Every requirement against the calculations. The status is met, not met or not verifiable at TRL 3; R12 is reported against the value-engineering target.

ID	Requirement	Target	Value	Status
R3	Climb speed	17 steps/min or more at rated load	260 W peak on a 250 W motor; 16.3 steps/min on that motor	Not met
R5	Truck mass	27 kg or less; pack 3 kg or less	34.2 kg; pack 2.6 kg	Not met
R6	Operator handle force	100 N or less inside the tilt window	121 N at the plus or minus 6 degree edge; 76 N at the set angle	Not met
R1	Rated stair load	60 kg up and down; 100 kg on the flat	Safety factors 1.4 shaft, 1.5 spider, 2.6 rail at 3 g	Met
R2	Stair range	Risers 100 to 200 mm, treads 250 mm or more, nosing up to 32 mm	Landing 66 mm past the nosing at 200 mm; tread needed 240 mm; arms clear 32 mm overhangs	Met
R4	Endurance	1,000 loaded steps up plus 1,000 down	1,215	Met
R10	Flat rolling	40 N or less	28 N	Met
R11	Size	Width 600 mm or less; upright height 1,500 mm or less	588 mm over the wheel end screws; 1,451 mm	Met
R13	Battery	24 V LiFePO4, BMS, fused, 0 to 45 °C charge window, 4 h charge	3.5 h	Met
R15	Stair and building protection	No steel contact with stairs; skids over nosings	Shaft-line guard 53.9 mm against 54.2 mm allowed; countershaft guard clears by 80 mm; every frame-fixed outline 10 mm or more from every nosing	Met
R12	Affordable	Value-engineering target USD 650	USD 776	USD 126 over the target
R7	Tilt control	100 Hz IMU, plus or minus 6 degree window, stop within 0.2 s	Control concept only	Not verifiable at TRL 3
R8	Hold on any loss	Holds with power off; drift 5 mm or less in 10 min	Worm self-locking; brake margin 7.5 times	Not verifiable at TRL 3
R9	Hold-to-run	Stops within 0.2 s of grip release	Circuit concept only	Not verifiable at TRL 3
R14	Environment	0 to 40 °C, IP54, rain on stoops	Enclosure concept only	Not verifiable at TRL 3

Counts: 7 met, 3 not met, 4 not verifiable at TRL 3, and R12 reported against the value-engineering target.

■ Safety

Safety: These are paper numbers for a 94 kg load moving on a stair with a person uphill of it. The design keeps steel parts clear of nosings on paper, but the chain case sits at the limit of its envelope on the shaft line, so a bent case or a nosing outside the R2 range could still be struck; a strike mid-step could jerk the frame toward the courier or stall the climb with the load half-lifted. The grip force at the window edge (121 N) is above target. The holding figures in section 6 assume a static load; worm creep under vibration and the brake's real holding torque must be confirmed before anyone climbs with a load. The 3 g shock factor is a judgment, not a measurement.

■ References

- SCM-REQ-001 v0.5, requirements.
- SCM-PRC-001 v0.5, design precis.
- SCM-DDR-001, TRL 2 review decisions; SCM-DDR-002, recommendations accepted; SCM-DDR-003, design for construction.
- SCM-BLD-001, prototype build plan; SCM-DEC-001, design decisions register.
- Flange bearing, weld-on hub and sprocket masses: catalogue-class values for 20 and 25 mm two-bolt flange units, 1610 weld-on hubs and plate sprockets, to confirm when bought.
- ISO 606, short-pitch transmission precision roller chains (minimum tensile strengths, confirm with the chain supplier's catalog).
- US International Residential Code, stair riser and tread limits, as summarized in SCM-PRB-001.